

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1939

South Meadow 345 kV
May 28, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1939. The Interconnection Customer proposes a 396.5 MW battery energy storage facility interconnecting at the South Meadow 345 kV substation in the Pennsylvania area. The request was received on November 25, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 567 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,853,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: CEDAR CRK / TARPON SPRINGS 138kV
Contingency: CEDAR CRK / YORKANA 345kV
Loading: 103.9% (Rating: 375 MVA)
DFAX: 5.9%
Pre-project loading: 97.6%
- Facility: S MEADOW / POCKET NORTH 345kV L
Contingency: SOUTH MEADOW / MONROE NE 345kV Line
Loading: 102.9% (Rating: 1,261 MVA)
DFAX: 50.5%
Pre-project loading: 87.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CEDAR CRK / TARPON SPRINGS 138kV	Reconductor line	\$378,000	20% (\$77,000)
S MEADOW / POCKET NORTH 345kV L	Reconductor line	\$8,828,000	20% (\$1,776,000)
	Total allocated to project		\$1,853,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.